AKHILENDRA DWIVEDI

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PROFESSIONAL SUMMARY

M.Tech AI Candidate at NIT Jalandhar with a background in full-stack web application development, microservices, and Retrieval-Augmented Generation (RAG) pipelines. Currently conducting academic research in Wi-Fi Sensing, processing high-dimensional signal analytics and sensor datasets for machine learning applications.

EDUCATION

Master of Technology in Artificial Intelligence — Aug 2025 - June 2027
Dr B R Ambedkar National Institute of Technology (NIT) Jalandhar, Punjab — CGPA: 8.29/10.00

Bachelor of Technology in Information Technology — Nov 2020 - June 2024
Dr Ambedkar Institute of Technology for Handicapped (AKTU), Kanpur, UP — CGPA: 8.06/10.00

SKILLS

Languages: Python, JavaScript, SQL, C++, C
AI & Machine Learning: Deep Learning, NLP, LLM Fine-Tuning, Embeddings, Scikit-Learn
Agentic Pipelines: RAG, Multi-Agent Frameworks
Databases: Neo4j, Qdrant, ChromaDB, MongoDB Atlas, MySQL
Backend & Frameworks: FastAPI, Node.js, Express.js, React.js, Vite, Tailwind CSS
Developer Tools & Core CS: Docker, Docker Compose, Git, Data Structures & Algorithms, OOPs

PROJECTS

Gov Scheme AI – AI-Powered Recommendation Platform — June 2026 - Present

- Engineered a hybrid retrieval system matching public policies via dense **semantic search** (**Qdrant**) and rule-based filters.
- Built high-performance asynchronous microservices using **FastAPI** and **Node.js/Express** to handle natural language inputs.
- Integrated **OpenRouter model** for multilingual generation and containerized the architecture using **Docker** and **Docker Compose**.

Narrative Consistency Detection, KDSH Finalist, IIT Kharagpur x Pathway — Dec 2025 - Jan 2026

- Architected a **RAG** reasoning pipeline utilizing **Pathway ETL** for stream data processing and text anomaly detection.
- Deployed open-weight LLMs (**Qwen2.5-7B**) locally via **Ollama** to run complex contextual logic verification on text datasets.
- Utilized **Sentence Transformers** (`all-mpnet-base-v2`), **Scikit-Learn**, and **Pandas** to compute semantic similarity thresholds.

LitNavigator AI – Multi-Agent Research Intelligence Platform — March 2026 - April 2026

- Built an autonomous research engine using **LangChain** multi-agent architectures to orchestrate recursive literature reviews.
- Designed a hybrid retrieval framework leveraging **ChromaDB** for dense vector lookups and **BM25** for sparse keyword matching.
- Constructed **Knowledge Graphs** in **Neo4j** to map citation dependencies, evaluating agent streams with **LangSmith**.